

MSc(PG) Introduction to Materials Modelling

Past Test Question

Feb 2026

Here are a few questions from past test. This is not a complete past test, but just a few examples of questions. This is meant so that you have an idea of what to expect in the test. There can question on topics covered in the lectures/practicals but not covered in the example question given below.

A test can have a total of 60 Marks

Example 1:

‘Every model captures specific aspects of reality’. Explain strengths and weaknesses of models in general and select one example to illustrate your explanation. **[5 Marks]**

Example 2:

How does the characteristic time scale models change when the characteristic length scale of a increases? **[3 Marks]**

Example 3:

Discuss the statement: “Calphad calculations, and in particular Thermo-Calc, can only do calculations for up to three components. This is the main limitation of the Calphad method.”

[5 Marks]

Example 4:

You are asked to perform Thermo-Calc calculations of a stainless steel Fe-18Cr-8Ni-0.1C, all wt%.

- i.) Which database do you used? Explain why this one and how to check the database is suitable for the calculation. **[3 Marks]**
- ii.) Which steps are you taking to perform this calculation? **[3 Marks]**
- iii.) Which phases are stable at 1000 C? What there composition? Is any of the phases ordered? **[4 Marks]**

- iv.) Calculate which phases appear at different temperatures and determine the equilibrium temperatures. Upload the corresponding diagram, should there be one.
- [5 Marks]**

For other questions you should be familiar with

- the equations for Gibbs energy
- basics of different modelling methods
- the simple model implemented in excel at the start of the module.
- How to do diffusion simulations in Thermo-Calc